Advanced Reinforcement Learning Strategies for Multi-Environment UGV Navigation

Phase II: Dynamic Scenarios and Liquid Time-constant Integration

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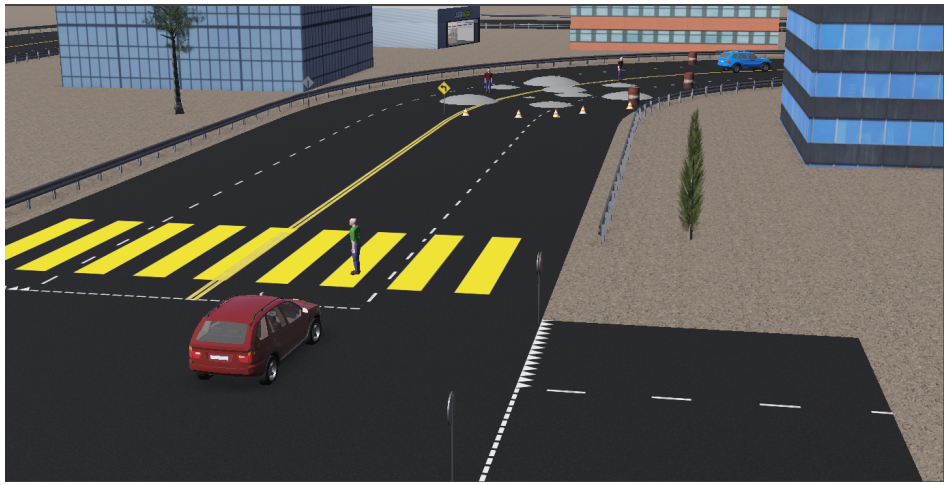
Project Progression: Phase I to Phase II

- **Phase I Success:** Established a baseline comparison between Dueling DQN and SAC in a simplified perception framework.
- **Phase II Objective:** Transition from static, simple environments to **complex dynamic road networks** and **diverse static terrains**.
- **Architectural Innovation:** Replacing standard LSTMs with LTCs-based neural architectures to handle temporal irregularities.

1. Complex Plain Road Dynamic Environment

- **Simulation Engine:** Webots 2025a.
- **Dynamic Obstacles:** Integrated moving vehicles and pedestrian agents with randomized velocities.
- **Spatial Complexity:** Multi-lane road layout with intersections to test yield behavior.

(Dynamic Environment Simulation)



2. Comparative Performance Analysis

- **DQN (Discrete):** Faced significant challenges in dynamic environments. The discrete action grid led to a high **Collision Rate (51.3%)** due to jittery maneuvers in tight spaces.
- **SAC (Continuous):** Achieved superior results with a **Success Rate of 71.6%**. The continuous action space allowed for precise, fluid swerving.
- **Finding:** SAC's entropy-regularized policy effectively halved the collision rate (**24.6%**) compared to the DQN baseline.

3. Replacing LSTM with Liquid Time-constant (LTC) Networks

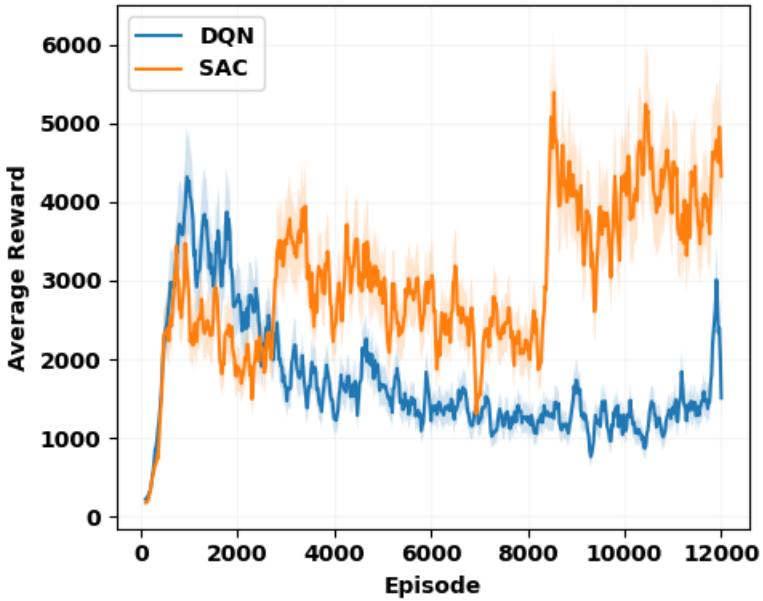
- **The Problem:** Standard LSTMs assume fixed time steps, which fails when LIDAR data or simulator frames drop/lag.
- **The Solution:** Implemented **LTCs**, which model hidden states via ordinary differential equations (ODEs).
- **Impact:** Improved temporal stability. The vehicle maintains "memory" of obstacle trajectories even under irregular sampling rates.

$$\frac{dx}{dt} = -[w_{leak} + \sum f_i(x, u)] \odot x + \sum f_i(x, u) \odot A_i$$

4. Training Evolution: Average Reward

- **SAC (Orange):** SAC maintained a stable reward level through the mid-training phase. SAC agent's average reward surged at 8,500 episode and stabilized at a high plateau between 4,000 and 5,500.
- **DQN (Blue):** This agent exhibited a rapid and aggressive initial learning phase, reaching a peak average reward of approximately 4,300 within the first 1,000 episodes then performance degraded sharply after episode 2,000.
- **Convergence:** The surge in SAC's performance after episode 8,500 suggests a successful transition from exploration to an optimal exploitation.

(Average Reward vs Episode)



5. Tailored Reward Function (Dynamic Environment)

The reward function was updated to penalize "Near-Misses":

Dynamic Reward Formulation

$$R_{total} = r_{goal} + r_{col} + r_{head} + \mathbf{r_{TTC}}$$

- **Time-to-Collision (TTC):** A weighted penalty based on the closing speed between the UGV and moving obstacles.
- **Smoothness Penalty:** To reduce jittering caused by high-frequency velocity changes in the dynamic environment.

6. Quantitative Results: Dynamic Navigation

Table: Performance Summary: DQN vs SAC

Metric	**Dueling DQN**	**SAC (LTC)**
Success Rate (%)	39.2%	**71.6%**
Collision Rate (%)	51.3%	**24.6%**
Avg. Convergence (Ep)	∼12,000	∼**8,000**

- SAC demonstrates a **nearly 2x improvement** in success rate.
- Implementation of LTCs and continuous control reduced collisions by **52%** relative to the DQN baseline.

7. Identified Limitations & Future Work

- **Sensory Sparsity:** The 32-bin LIDAR descriptor occasionally misses very thin obstacles in complex terrains.
- **Computational Overhead:** LTC networks require more backpropagation time due to ODE solving.
- **Currently Working on:** High speed vehicle and Sensor Fusion.
- **Future Work:** Transitioning the LTC-SAC weights onto a physical Raspberry Pi-based UGV platform for real-world validation.